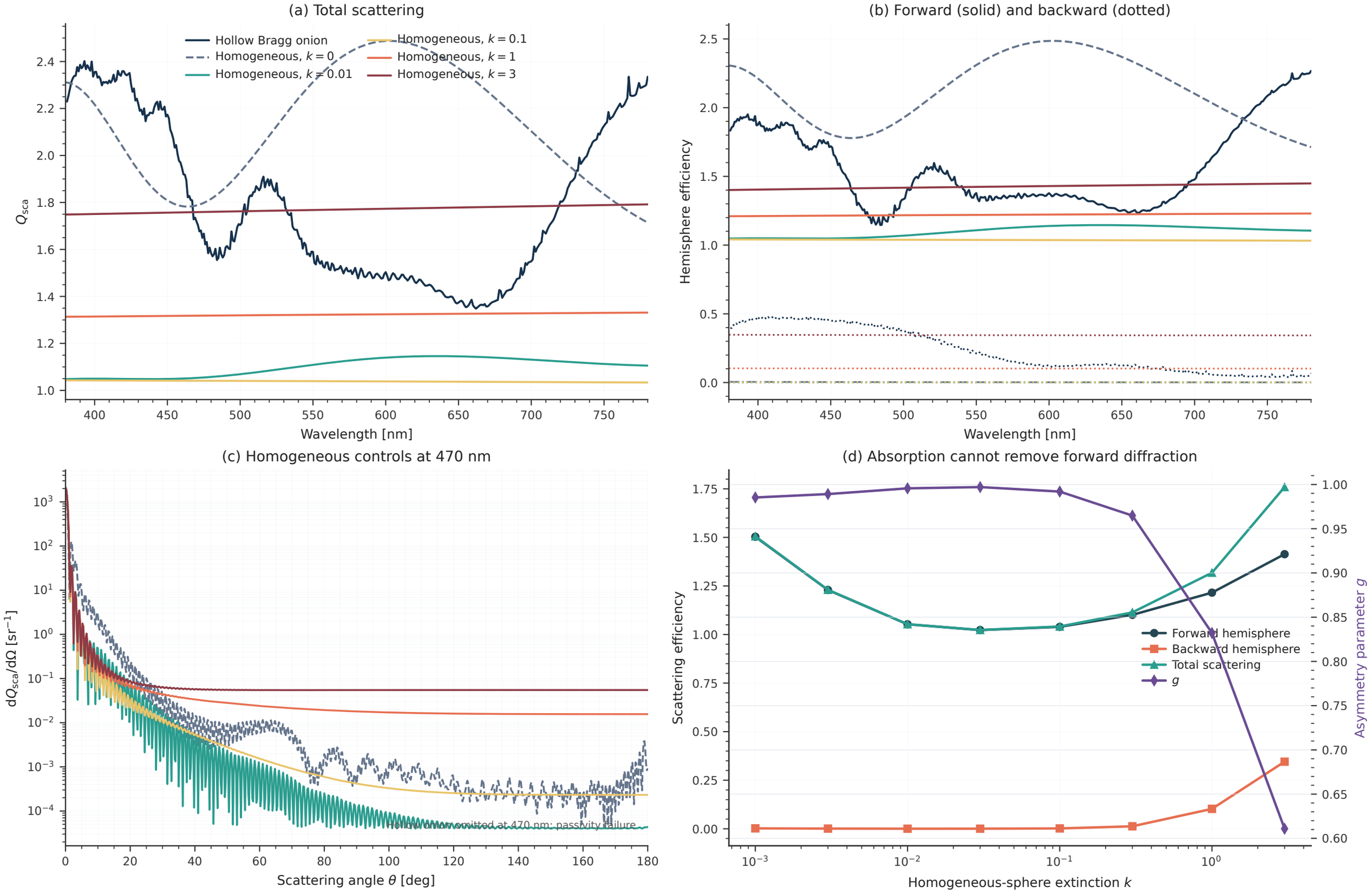

(a) Total scattering
Hollow Bragg onion
Homogeneous, $k = 0$
Homogeneous, $k = 0.01$
Homogeneous, $k = 0.1$
Homogeneous, $k = 1$
Homogeneous, $k = 3$
Q_{sca}
Wavelength [nm]
(b) Forward (solid) and backward (dotted)
Hemisphere efficiency
Wavelength [nm]
(c) Homogeneous controls at 470 nm
$dQ_{sca}/d\Omega$ [sr^{-1}]
Hollow shell committed at 470 nm: passivity failure
Scattering angle θ [deg]
(d) Absorption cannot remove forward diffraction
Scattering efficiency
Asymmetry parameter g
Forward hemisphere
Backward hemisphere
Total scattering
g
Homogeneous-sphere extinction k